

GIUSEPPE GERARDI

Senior Risk Manager | IRRBB & Liquidity Risk | Quantitative Analyst

Geneva, Switzerland | +41 782 623 071 | giu.gerardi@gmail.com | linkedin.com/in/giuseppegerardi

PROFESSIONAL SUMMARY

Senior Risk Manager with 7+ years of international experience, of which 3+ in banking (**REYL Intesa Sanpaolo**, Geneva). Specialist in financial risk modelling with deep expertise in **IRRBB, LCR/NSFR and regulatory stress testing** under **FINMA and ECB** frameworks, including end-to-end data integration from core banking (Finnova) into **Prometeia ERMAS** with full second-level control over regulatory data flows. Reduced reporting production cycles from **several days to a few hours** by engineering automated pipelines in Python, R and SQL — an approach now extended toward **Agentic AI and RAG architectures** to shift risk monitoring from reactive to predictive and regulatory documentation from manual to intelligent.

TECHNICAL SKILLS

Programming	Python (Pandas, NumPy, SQLAlchemy), R (Shiny, tidyverse), Git / GitHub
Data & Databases	Advanced SQL (PostgreSQL, SSMS), NoSQL (MongoDB), Neo4j, ETL/ELT, Informatica PowerCenter
Risk Modelling	IRRBB, LCR, NSFR, VaR, Stressed VaR, IRC, Expected Shortfall, Monte Carlo, Stress Testing, Backtesting
Machine Learning	Supervised & Unsupervised, Gradient Boosting (XGBoost), NLP, Transformers, RNN/LSTM, CNN
Agentic AI	LangGraph, LangChain, RAG (ChromaDB, FAISS), Prompt Engineering, Chain-of-Thought (CoT)
RegTech & Systems	Finnova (Core Banking), Prometeia ERMAS (ALM/Risk), Regnology (FiRE), FINMA, ECB, Basel III / CRR, EBA IRRBB Guidelines

CORE COMPETENCIES

✓ IRRBB & Liquidity Risk (LCR, NSFR)	✓ Agentic AI & RAG for Predictive Risk Monitoring
✓ Stress Testing & Scenario Analysis (FINMA, ECB)	✓ Intelligent Regulatory Documentation (LLM)
✓ Regulatory Reporting Automation (Python, R, SQL)	✓ Quantitative Analytics & Statistical Modelling
✓ Data Governance & Regulatory Data Quality	✓ Regulatory Compliance (FINMA, ECB, Basel III)

PROFESSIONAL EXPERIENCE

Senior Risk Manager — Intesa Sanpaolo Wealth Management Suisse

Geneva, CH | Jan 2023 – Present

Ownership & Responsibilities

- **Risk Models:** Owner of 3+ quantitative risk models (IRRBB, LCR, NSFR, parametric VaR, Stressed VaR, IRC, Expected Shortfall, stochastic scenario analysis) — responsible for theoretical design, data modelling, technical implementation and quarterly backtesting.
- **ICS & Compliance:** Lead assessor of the Internal Control System across market risk and banking book processes; identified and closed regulatory control gaps (FINMA Circ. 2017/1, Basel III, EBA IRRBB Guidelines) and maintained continuous alignment with FINMA and ECB regulatory developments.
- **FINMA Data Governance:** Redesigned the data lineage for FINMA 08/2024 regulatory reporting on PostgreSQL, enabling automatic reconciliation between front-office positions (Finnova) and regulatory aggregates with an error tolerance of < 0.01%.

- **Data Engineering & Second-Level Control:** Responsible for the data engineering integration from core banking (Finnova) into Prometeia ERMAS, with full ownership of second-level controls over regulatory data flows — ensuring integrity, completeness and traceability of all inputs feeding IRRBB and liquidity models.

Key Achievements — Risk ∩ Automation

- **Reporting Automation:** Migrated the full regulatory reporting cycle (stress testing, IRRBB, LCR/NSFR) from manual Excel/VBA processes to end-to-end pipelines in R, Python and SQL — cutting processing time from several days to a few hours and releasing analytical capacity for forward-looking scenario work.
- **Interactive Simulation & Modeling Tools:** Built interactive simulation engines and analytical tools (Python/R Shiny deployed via Posit Connect) that simplified how internal teams interact with complex financial data-enabling decision-makers to run scenario analyses on interest rate and liquidity shocks and make faster, data-driven decisions.
- **Streamlined Regulatory Reporting & Documentation:** Optimized the regulatory reporting workflow by transforming manual, document-based processes into structured, automated pipelines-drastically reducing drafting friction, ensuring full terminological consistency, and elevating the overall quality and reliability of regulatory submissions.
- **Data Accessibility & Process Automation:** Spearheaded the development of internal web-based workflow and monitoring tools that bridge the gap between risk management and data flows; by replacing unstructured inputs with clean, standardized data models, the initiative eliminated manual follow-ups, streamlined cross-departmental collaboration, and built a solid foundation for robust data governance.
- **Finnova → Prometeia ERMAS Pipeline:** Designed and implemented the data engineering pipeline integrating core banking data (Finnova) into Prometeia ERMAS — ensuring quality and consistency of regulatory inputs for IRRBB and liquidity calculations. The second-level control framework systematised data validation, reducing feed anomalies and increasing the reliability of FINMA regulatory outputs.

Data Scientist — Città Metropolitana di Milano

Milan, IT | May 2021 – Dec 2022

- Replaced 10+ manual Excel workflows with Python ETL pipelines (Pandas, SQLAlchemy), shifting reporting frequency from monthly to weekly — the same automation methodology later applied to regulatory reporting at REYL.
- Translated institutional stakeholder requirements into 3 BI models and interactive dashboards (Python / Tableau) adopted by the Metropolitan Council for budget and planning decisions.
- Designed a data quality framework for high-variability public datasets (Open Data), ensuring the integrity of statistical aggregates used in official reports — methodological foundation for subsequent FINMA data governance work.

Data Manager & Quantitative Analyst — Università degli Studi di Milano

Milan, IT | Oct 2019 – Apr 2021

- Data Management & Data Quality Governance: Designed, audited, and refactored end-to-end data pipelines for EU-funded research projects; established strict data validation and cleaning protocols that corrected systematic errors, eliminated statistical bias, and guaranteed data integrity, completeness, and traceability for peer-reviewed publication.
- HPC Infrastructure & Cross-Departmental Collaboration: Spearheaded strategic internal initiatives to upgrade computational facilities, successfully driving the collaboration between the department and UNIMI INDACO (High-Performance Computing Platform) to expand processing power and empower researchers with advanced, scalable experimental environments.
- Scientific Data Architecture: Managed the full lifecycle of complex research datasets, implementing robust quantitative architectures (Python/Django, R) to structure, standardize, and securely store multi-source data supporting international scientific publications.
- COVID-Pro Project & Epidemiological Modeling: Served as lead Data Manager for a national-scale epidemiological initiative, developing a province-based time-dependent SIRD model in Python to process live health datasets, generate real-time analytical dashboards, and translate complex health data into actionable insights for policymakers (Published in IJERPH, Impact Factor 4.6, 2021).

Data Engineering Consultant — SDG Group

Milan, IT | Nov 2018 – Sep 2019

- Developed enterprise-grade ETL pipelines (Informatica PowerCenter) on IBM Netezza architectures for financial sector clients, managing Big Data scale volumes — foundational experience for the regulatory pipelines built subsequently.
- Built NLP modules (IBM Watson NLU + Python) for automated anomaly ticket classification on ETL flows, reducing manual triage and accelerating time-to-resolution — first applied use of NLP techniques later evolved into LLM architectures.
- Managed production workflow monitoring and escalation via ServiceNow, ensuring SLA compliance across multi-client environments.

Data Scientist & Analyst — **UVET American Express GBT** *Milan, IT | Sep 2018 – Nov 2018*

- Built OCR pipelines (Python Tesseract + regex NLP) for automated structured-data extraction from unstructured travel documents — document extraction competency later evolved into RAG architectures for regulatory documentation.
- Conducted Sentiment Analysis on customer experience data (R/RStudio Server) to surface satisfaction drivers, delivering actionable insights to the commercial team.

SIDE PROJECTS / PERSONAL R&D

Quantitative Financial Data Pipeline & Agentic AI Architecture | *Ongoing R&D*

- **Financial Data Management & Ingestion Architecture:** Engineered a scalable data layer integrating OpenBB and multi-source financial APIs (FRED, SEC, Eurostat); built automated pipelines for point-in-time Parquet caching, geometric cleaning, and zero-leakage feature processing yielding 80+ atomic scalar metrics.
- **Agentic AI & Multi-Model Data Integration:** Implemented an agentic analytics layer combining local Edge LLMs (Ollama) and cloud APIs to dynamically query, structure, and synthesize unstructured market feeds, financial news, and sentiment data (FinBERT).
- **Deterministic-to-AI Pipeline Design:** Established a strict architectural separation between the deterministic quantitative calculation engine and agentic reasoning modules, optimizing context windows and ensuring clean, reproducible structured data handoffs for AI-driven portfolio intelligence.

EDUCATION

MSc in AI & AI Agents for Business *May 2026 - Present (Expected 2027)*

Università UniMarconi Torino (in collaboration with start2impact)

Accredited Executive Master focused on advanced AI architectures and hands-on business automation (27 real-world enterprise projects).

- *AI & Machine Learning Focus: Deep dive into AI Business Strategy, prompt engineering, Retrieval-Augmented Generation (RAG), and the deployment of autonomous AI Agents designed to automate operational workflows and risk monitoring.*
- *Data & Analytics Foundation: Advanced implementation of Machine Learning frameworks, data science architectures (Python, SQL, NumPy), and advanced data visualization (Tableau).*

MSc in Data Science for Economics, Business and Finance *2018*

Università degli Studi di Milano

Machine Learning & Data Mining, Computational Statistics, Financial Data Science, AI & Expert Systems, Social Network & Sentiment Analysis.

MSc in Business Administration and Management *2013 | 102/110*

Università degli Studi di Roma Tre

BSc in Business Economics *2011*

Università degli Studi della Tuscia | Erasmus Exchange: Universität Augsburg, Germany

CERTIFICATIONS & CONTINUOUS LEARNING

Business Analytics in the Age of Generative AI — London Business School

ACADEMIC PUBLICATIONS

Modeling Provincial COVID-19 Epidemic Data Using an Adjusted Time-Dependent SIRD Model | International Journal of Environmental Research and Public Health (IF 4.6) | Jun 2021

Analyzing the Waiting Time of Academic Publications: A Survival Model | Pearson – 50th Scientific Meeting of the Italian Statistical Society | Jul 2020

LANGUAGES

Italian Native **English** C1 (Professional) **French** B1 (Working environment, Geneva)